Q1 - 24 June - Shift 1

A plane electromagnetic wave travels in a medium of relative permeability 1.61 and relative permittivity 6.44. If magnitude of magnetic intensity is 4.5×10^{-2} Am^{-1} at a point, what will be the approximate magnitude of electric field intensity at that point ?

(Given : permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}, speed of light in vacuum $c = 3 \times 10^8$ ms^{-1})

(A) 16.96 Vm^{-1} (B) 2.25×10^{-2} Vm^{-1}

(C) 8.48 Vm^{-1} (D) 6.75×10^6 Vm^{-1}

Space for your notes:

Q2 - 25 June - Shift 1

The electric field in an electromagnetic wave is given by $E = 56.5 \sin \omega(t - x/c) NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space. (Given $\varepsilon_0 = 8.85 \times 10^{-12} C^2N^{-1}m^{-2}$)

(A) 5.65 Wm^{-2} (B) 4.24 Wm^{-2}

(C) 1.9×10^{-7} Wm^{-2} (D) 56.5 Wm^{-2}

Space for your notes:

Q3 - 25 June - Shift 2

The electromagnetic waves travel in a medium at a speed of 2.0×10^8 m/s.The relative permeability of the medium is 1.0. The relative permittivity of the medium will be:

(A) 2.25 (B) 4.25

(C) 6.25 (D) 8.25

Space for your notes:

Q4 - 26 June - Shift 1

If electric field intensity of a uniform plane electro magnetic wave is given as

$E = -301.6 \sin(kz - \omega t)\hat{a}_x + 452.4 \sin(kz - \omega t)\hat{a}_y \frac{V}{m}$

Then, magnetic intensity H of this wave in Am^{-1} will be:'

[Given: Speed of light in vacuum $c = 3 \times 10^8\ ms^{-1}$, permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$]

(A) $+0.8\sin(kz-\omega t)\hat{a}_y + 0.8\sin(kz-\omega t)\hat{a}_x$

(B) $+1.0\times10^{-6}\sin(kz-\omega t)\hat{a}_y + 1.5\times10^{-6}(kz-\omega t)\hat{a}_x$

(C) $-0.8\sin(kz-\omega t)\hat{a}_y - 1.2\sin(kz-\omega t)\hat{a}_x$

(D) $-1.0\times10^{-6}\sin(kz-\omega t)\hat{a}_y - 1.5\times10^{-6}\sin(kz-\omega t)\hat{a}_x$

Space for your notes:

Q5 - 26 June - Shift 2

Which is the correct ascending order of wavelengths?

(A) $\lambda_{\text{visible}} < \lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{microwave}}$

(B) $\lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

(C) $\lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

(D) $\lambda_{\text{microwave}} < \lambda_{\text{visible}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}}$

Space for your notes:

Q6 - 27 June - Shift 1

Mach List-I with List – II :

List – I List – II

	List-I		List-Ii
(a)	Ultraviolet rays	(i)	Study crystal structure
(b)	Microwaves	(ii)	Greenhouse effect
(c)	Infrared waves	(iii)	Sterilizing surgical instrument
(d)	X-rays	(iv)	Radar system

Choose the correct answer from the options given below :

(A) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(B) (a) – (iii), (b) – (i), (c) – (ii), (d) – (iv)

(C) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(D) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

Space for your notes:

Q7 - 27 June - Shift 2

Given below are two statements:

Space for your notes:

Statement I : A time varying electric field is a source of changing magnetic field and vice-versa. Thus a disturbance in electric or magnetic field creates EM waves.

Statement II : In a material medium. The EM wave travels with speed $v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both statement I and statement II are true.

(B) Both statement I and statement II are false.

(C) Statement I is correct but statement II is false.

(D) Statement I is incorrect but statement II is true.

Q8 - 28 June - Shift 2

An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :

(A) $E_y = 60 \sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8 t\right)\right]\hat{j}\,Vm^{-1}$

$B_z = 2\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8 t\right)\right]\hat{k}\,T$

(B) $E_y = 60 \sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8 t\right)\right]\hat{j}\,Vm^{-1}$

$B_z = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8 t\right)\right]\hat{k}\,T$

(C) $E_y = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8 t\right)\right]\hat{j}\,Vm^{-1}$

$B_z = 60\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8 t\right)\right]\hat{k}\,T$

(D) $E_y = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^4\left(x - 4\times 10^8 t\right)\right]\hat{j}\,Vm^{-1}$

$B_z = 60\sin\left[\frac{\pi}{4}\times 10^4\left(x - 4\times 10^8 t\right)\right]\hat{k}\,T$

Space for your notes:

Q9 - 29 June - Shift 1

The intensity of the light from a bulb incident on a surface is 0.22 W/m^2. The amplitude of the magnetic field in this light-wave is______$\times 10^{-9}$ T.

(Given : Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$, speed of light in vacuum $c = 3 \times 10^8 ms^{-1}$)

Space for your notes:

Answer Key

Q1 (C)	**Q2** (B)	**Q3** (A)	**Q4** (C)
Q5 (B)	**Q6** (A)	**Q7** (C)	**Q8** (B)
Q9 (43)			

Q1 (C)

$\mu_r = 1.61 \qquad \in_r = 6.44$

$B = 4.5 \times 10^{-2}$

$E = ?$

$C = \frac{1}{\sqrt{\mu_0 \in_0}} \quad V = \frac{1}{\sqrt{\mu \in}}$

$\frac{C}{V} = \sqrt{\mu_r \in_r} = \sqrt{1.61 \times 6.44}$

$\frac{E}{B} = V = \frac{3 \times 10^8}{\sqrt{1.61 \times 6.44}} = 9.32 \times 10^7 \text{ m/s}$

$E = 4.5 \times 10^{-2} \times 9.32 \times 10^7$

$= 4.2 \times 10^6$

Q2 (B)

$I = \frac{1}{2} \varepsilon_0 E_0^2 c$

$I = \frac{1}{2} \times (8.85 \times 10^{-12})(56.5)^2 \times (3 \times 10^8)$

$= 4.24 \text{ Wm}^{-2}$.

Q3 (A)

$$X_L = 10^{-2} \times 3000 = 30\,\Omega$$

$$X_C = \frac{1}{3000 \times 25 \times 10^{-6}} = \frac{40}{3}\,\Omega$$

$$X = X_L - X_C$$

$$= 30 - \frac{40}{3} = \frac{50}{3}$$

$$\tan\delta = \frac{X}{R} = \frac{50}{3 \times 100} = \frac{1}{6}$$

$$\delta = \tan^{-1}\left(\frac{1}{6}\right) = \tan^{-1}(0.17)$$

Q4 (C)

$\vec{E} = 301.6\sin(kz - \omega t)(-\hat{a}_x) + 452.4\sin(kz - \omega t)\hat{a}_y$

$\vec{B} = \frac{301.6}{C}\sin(kz - \omega t)(-\hat{a}_y)$

$+\frac{452.4}{C}\sin(kz - \omega t)(-\hat{a}_x)$

$\vec{H} = \frac{\vec{B}}{\mu_0} = \frac{301.6}{\mu C}\sin(kz - \omega t)(-\hat{a}_y)$

$+\frac{452.4}{\mu C}\sin(kz - \omega t)(-\hat{a}_x)$

$\vec{H} = -0.8\sin(kz - \omega t)\hat{a}_y - 1.2\sin(kz - \omega t)\hat{a}_x$

For direction

$\vec{E} \times \vec{B}$ is direction of $\vec{C}$

For first part $\hat{E} = -\hat{i}, \hat{B} = ?$

$\hat{E} \times \hat{B} = \hat{k} \Rightarrow \hat{B} = -\hat{j}$

Similarly for second

$\hat{E} = \hat{j}, \hat{B} = ?$

$\hat{E} \times \hat{B} = \hat{k} \Rightarrow \hat{B} = -\hat{i}$

Q5 (B)

From electromagnetic wave spectrum.

λ increases $\longrightarrow$

γ-ray	x-rays	ultra violet	visible	infrared	microwave	Radio wave

$\lambda_{\text{gamma-ray}} < \lambda_{X\text{-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

Q6 (A)

$$k=\frac{P^2}{2m} \Rightarrow P \alpha \sqrt{m}$$

$$\text{Now } \lambda=\frac{h}{p}$$

$$\text{So, } \lambda \alpha \frac{1}{p} \Rightarrow \lambda \ \alpha \ \frac{1}{\sqrt{m}}$$

$$\frac{\lambda_\alpha}{\lambda_{C12}}=\frac{\sqrt{3}}{1}$$

Q7 (C)

The statement II is wrong as the velocity of εm wave in a medium is $\frac{1}{\sqrt{\mu\varepsilon}}=\frac{1}{\sqrt{\mu_0\mu_r\varepsilon_0\varepsilon_r}}$.

Q8 (B)

$$B_0 = \frac{E_0}{c} = \frac{60}{3\times10^8} = 2\times10^{-7}\,T$$

$\hat{E}\times\hat{B}$ must be direction of propagation.

So, $\hat{B}\rightarrow$ z-axis

$$k = \frac{2\pi}{\lambda} = \frac{\pi}{4}\times10^3\,m^{-1}$$

$$E_y = 60\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{j}\,Vm^{-1}$$

$$B_z = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{k}\,T$$

Q9 (43)

$$I = \left(\frac{1}{2}\varepsilon_0 E_0^2\right)C$$

$$\Rightarrow E_0 \Rightarrow \sqrt{\frac{2I}{\varepsilon_0 C}} \Rightarrow \sqrt{\frac{2\times0.22}{8.85\times10^{-12}\times3\times10^8}} = 12.873$$

$$B \Rightarrow \frac{E_0}{C} \Rightarrow \frac{12.873}{3\times10^8} = 4.291\times10^{-8} = 43\times10^{-9}$$